

# Parent Guide — middle / module-12- movement-of-sun

IKS Curriculum

## Table of Contents

## Parent Guide — Module 12: Movement of the Sun

**Your child's age:** Middle (Grades 6–8) · **Module length:** 10 sessions × 45 min over ~2 weeks

### What your child is learning

Where the Sun is in the sky over the course of a year, and why. The module covers:

| Concept                                   | What it means                                                      |
|-------------------------------------------|--------------------------------------------------------------------|
| The ecliptic ( <i>krānti-vṛtta</i> )      | The yearly path the Sun <i>appears</i> to trace against the stars  |
| Axial tilt (~23.5°)                       | Why we have seasons                                                |
| <i>Uttarāyaṇa</i> / <i>Dakṣiṇāyaṇa</i>    | The Sun's northward / southward half-year courses                  |
| Solstices and equinoxes ( <i>viṣuva</i> ) | The four turning-points of the year                                |
| The 12 <i>rāśis</i>                       | Meṣa, Vṛṣabha, Mithuna, ..., Mīna — the zodiac signs               |
| <i>Samkrānti</i>                          | The day the Sun moves into a new <i>rāśi</i> (12 times a year)     |
| <i>Sāyana</i> vs <i>Nirayana</i>          | The two different reference frames Indian and modern astronomy use |

### How we teach it

The framing is **astronomy first**. We lead with what your child can observe — sunrise direction, shadow lengths, the longest and shortest days — and then introduce the Indian terminology as a vocabulary for what was observed. We do NOT teach astrology (the prediction of personality or fortune from star positions). We DO teach the astronomy on which the Indian calendar is based.

NEP 2020 §4.6 (Integration of Indian Knowledge Systems with mainstream science) supports this approach.

### Homework you'll see

- Observe sunrise direction one morning this week (Day 1–2). No equipment needed.
- A shadow-length measurement at noon, using a vertical stick at home (Day 2–3).
- A “calendar marking” task — annotate a printed year-calendar with solstices, equinoxes, and *saṃkrāntis* (Day 4–5).

- A *rāśi*-wheel project — build a paper wheel showing the 12 *rāśis* and the approximate date the Sun is in each (Days 8–10).
- 6–8 writing prompts across the module.

## How to help

**Helpful:** - Go outside with your child one Sunday morning and notice where the Sun rises. Note that the spot will be a little different in three months. - If you have a *pañcāṅga* (traditional Indian almanac) at home, show it. Point out the *saṁkrānti* dates — they’re listed. - Discuss the festival days that depend on solar position: *Makara Saṁkrānti* / Pongal / Lohri / Bihu (~14 January) and *Karka Saṁkrānti* (~16 July). These are the same festival across different regional names. - Discuss the difference between the equinox (around 21 March / 23 September) and the date when crops or seasons “change” locally — they are not always the same, and the gap tells us something about local geography.

**Not helpful:** - Insisting “astrology is true / astrology is nonsense.” Neither belongs in this module. We’re learning where the Sun is, not what that means for personality. - Doing the project for your child. The struggle with the *rāśi*-wheel is part of the learning. - Telling your child that *Makara Saṁkrānti* is “the winter solstice.” It’s close, but not exactly — and the gap is a real intellectual puzzle the child will work through on Day 7. Don’t pre-resolve it.

## Citation standard

By Middle band, projects include at least one reference to a real source. For this module the sources are the **Sūrya-siddhānta** (a classical Indian astronomy text) and the standard NCERT science / geography chapters listed in *sources .md*. Your child does not need to quote either — paraphrasing with a reference is enough.

This trains the habit: real references from real texts, not made-up citations.

## What success looks like

By end of module, your child can:

- Name all 12 *rāśis* in Sanskrit with the English / Greek equivalent (Meṣa = Aries, etc.).
- Explain why we have seasons, using a torch and a tilted globe (or sketches of one).
- Identify the dates of the two solstices and two equinoxes on a calendar.
- Explain what *Makara Saṁkrānti* marks astronomically.
- State the difference between *sāyana* (tropical) and *nirayana* (sidereal) zodiacs, and that the gap (*ayanāṁśa*) is about 24° today.
- Build a working paper *rāśi*-wheel and explain it for 3 minutes.

## On respect across traditions

This is a science module. Some families read *pañcāṅgas* devotionally; some don’t engage with traditional Indian astronomy at all. The classroom holds both: the astronomy is presented as a fact about the sky, the Indian terminology is presented as a historically-developed vocabulary for that fact, and any religious or astrological reading is left for families to handle as they choose.

## Want to go deeper?

Senior band (Grade 9–12) extends to **precession** (the ~25,800-year wobble of Earth’s axis that causes the *ayanāṁśa* drift), **Hipparchus** (the Greek astronomer who first measured it ~130 BCE), and the **gola** (spherical geometry) chapters of the *Sūrya-siddhānta* and *Āryabhaṭīya*.  
Preview: [curriculum/senior/module-12-movement-of-sun/README.md](#).

## Safety note

**Never look at the Sun directly** — not even for a second. The classroom observation uses shadow lengths from a vertical stick (a gnomon), which is the safe way. Please reinforce this at home if your child does any outdoor observation.

## 10-Day Lesson Plan

### Module 12 (Middle) — 10-Day Lesson Plan

#### At a glance

| Day | Session title               | Lesson file                                      | Activity / assessment                                      |
|-----|-----------------------------|--------------------------------------------------|------------------------------------------------------------|
| 1   | What is the Sun Doing?      | <a href="#">days/day-01-suns-path.md</a>         | Shadow-stick observation                                   |
| 2   | Why Earth has Seasons       | <a href="#">days/day-02-axial-tilt.md</a>        | Torch-globe demo + exit ticket                             |
| 3   | Uttarāyaṇa & Dakṣiṇāyana    | <a href="#">days/day-03-ayana.md</a>             | Calendar marking; formative quiz Part A                    |
| 4   | The Equinoxes (Viṣuva)      | <a href="#">days/day-04-equinox.md</a>           | Day-length graph                                           |
| 5   | The 12 Rāśis                | <a href="#">days/day-05-rashis.md</a>            | Rāśi-wheel introduction; formative quiz Part B             |
| 6   | Samkrānti                   | <a href="#">days/day-06-sankranti.md</a>         | <a href="#">activities/activity-02-makara-sankranti.md</a> |
| 7   | Sāyana vs Nirayana          | <a href="#">days/day-07-sayana-nirayana.md</a>   | Compare-contrast worksheet                                 |
| 8   | Project Session 1: Build    | <a href="#">days/day-08-project-session-1.md</a> | Project draft                                              |
| 9   | Project Session 2: Refine   | <a href="#">days/day-09-project-session-2.md</a> | Project draft                                              |
| 10  | Final Assessment + Showcase | <a href="#">days/day-10-final-assessment.md</a>  | <a href="#">quizzes/summative.md</a> + project             |

## Pacing notes

- Day 1's shadow-stick observation needs a sunny mid-morning window. If your timetable lands the lesson in early morning or after a rain, move the outdoor part to a later day and use the time for the explanation.
- The torch-and-globe demo (Day 2) works best with the room blackout-curtained. If your room doesn't black out, demonstrate the day before in a darker space and take photos to project.
- Day 7 (*sāyana* / *nirayana*) is intellectually the densest. Don't squeeze it.
- Project work (Days 8–9) assumes pair vs. individual is decided by Day 5.

## Suggested running theme

Open each class with the same prompt on the board: “**Where was the Sun this morning when you left for school?**” (Direction: east-ish — but more north of east in summer, more south of east in winter.) Two students share for 30 seconds each. By Day 10 the class will have a felt sense that the Sun does not rise from the exact same spot every day.

## Formative vs. summative weight

- Formative quizzes (Days 3 and 5) are **not graded** — they are diagnostic. Use them to decide who needs scaffolding on Day 7.
- Summative quiz (Day 10) — see rubric in `assessments/rubric.md`. Combined with project, total 40 marks.

## What students leave the module with

- A completed paper *rāśi*-wheel with their own labelling.
- The 12 Sanskrit *rāśi* names with English / Greek-derived equivalents.
- A diagram in their notebook showing axial tilt and the four turning points of the year.
- A felt sense that “what day is *Makara Saṁkrānti*?” and “what day is the winter solstice?” are answers to two slightly different questions.

## Homework

### Homework — Movement of the Sun (Middle)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Sun's Yearly Path** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Axial Tilt and the Seasons** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Uttarāyaṇa and Dakṣiṇāyana** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Equinoxes — Viṣuva** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Twelve Rāśis** · Write 2 sentences about today's concept. (~10 min)

6. **Day 6 — Saṁkrānti Moments** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Sāyana vs Nirayana** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Precession of the Equinoxes** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Cross-Cultural Solar Calendars** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Share and Assessment** · Write 2 sentences about today's concept. (~10 min)

## Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.